

STAT 639 - Homework 1

ISLR Chapter 2: Exercises 1, 2, 8, and 10

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August 27, 2026

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1 Exercise 1

For each case, determine whether a more flexible statistical learning method would generally be expected to perform better or worse than a less flexible method.

1.1 (a) Large number of observations, small number of predictors

A more flexible method should generally perform better.

When n is very large and p is small, there is enough data to estimate a flexible model without suffering too much from high variance. A flexible model can capture more complicated relationships between the predictors and response, so the reduction in bias may outweigh the increase in variance.

1.2 (b) Small number of observations, large number of predictors

A more flexible method should generally perform worse.

With small n and large p , a flexible method can easily fit random noise in the training data. This leads to overfitting and high variance. A less flexible method is generally safer in this setting.

1.3 (c) Highly non-linear relationship between predictors and response

A more flexible method should generally perform better.

If the true relationship is highly non-linear, a flexible model is better able to approximate that relationship. A simple model may have high bias because it cannot capture the true structure.

1.4 (d) Very high variance of the error terms

A more flexible method should generally perform worse.

When the irreducible noise is very large, a highly flexible method may begin fitting the noise rather than the underlying signal. This increases variance and can hurt test-set performance.

2 Exercise 2

For each situation, identify whether the problem is classification or regression, whether the main goal is inference or prediction, and identify n and p .

2.1 (a) CEO salary

The response is CEO salary, and the explanatory variables include company profit, number of employees, and industry.

- **Type:** Regression
- **Goal:** Inference
- n : 500 firms
- p : 3 predictors

This is a regression problem because CEO salary is quantitative. The purpose is to understand which factors are associated with CEO salary, so the primary goal is inference.

2.2 (b) Success of a new product

The response indicates whether a product is successful or unsuccessful. The predictors consist of price, marketing budget, competitors' price, and ten other variables.

- **Type:** Classification
- **Goal:** Prediction
- n : 20 products
- p : 13 predictors

This is classification because the response is qualitative. The goal is to predict whether the new product will succeed.

2.3 (c) Weekly change in the U.S. dollar

The response is the weekly percentage change in the U.S. dollar relative to the euro, using weekly percentage changes in the stock market and the company's market as predictors.

- **Type:** Regression
- **Goal:** Prediction
- *n*: 52 weeks
- *p*: 2 predictors

The response is quantitative, so this is regression. Since the goal is to forecast the weekly percentage change, the main goal is prediction.

3 Exercise 8 - College Data

This exercise uses the `College` data set described in ISLR Chapter 2.

3.1 (a) Read the data

The code below first looks for `College.csv` in the current working directory. If it is not present, it downloads the file from the book website.

```
college_file <- "College.csv"

if (!file.exists(college_file)) {
  download.file(
    "https://www.statlearning.com/s/College.csv",
    destfile = college_file,
    mode = "wb"
  )
}

college <- read.csv("College.csv", check.names = FALSE)
```

There are 777 colleges and universities. Before removing the institution-name column, the imported data contain 19 columns.

3.2 (b) Use college names as row names

```
# Inspect first column
rownames(college) <- college[, 1]
View(college)
```

After removing the institution-name column, the data contain 777 observations and 19 variables. The college names are now stored as row names rather than as a predictor.

3.3 (c-i) Numerical summary

```
summary(college)
```

##	Private	Apps	Accept
## Length:777	Length:777	Min. : 81	Min. : 72
## Class :character	Class :character	1st Qu.: 776	1st Qu.: 604
## Mode :character	Mode :character	Median : 1558	Median : 1110
##		Mean : 3002	Mean : 2019
##		3rd Qu.: 3624	3rd Qu.: 2424

```
##                               Max. :48094  Max. :26330
##      Enroll      Top10perc      Top25perc      F.Undergrad
##  Min.   : 35      Min.   : 1.00      Min.   : 9.0      Min.   : 139
## 1st Qu.: 242      1st Qu.:15.00      1st Qu.: 41.0     1st Qu.: 992
## Median : 434      Median :23.00      Median : 54.0     Median : 1707
## Mean   : 780      Mean   :27.56      Mean   : 55.8     Mean   : 3700
## 3rd Qu.: 902      3rd Qu.:35.00      3rd Qu.: 69.0     3rd Qu.: 4005
## Max.   :6392      Max.   :96.00      Max.   :100.0     Max.   :31643
##  P.Undergrad      Outstate      Room.Board      Books
##  Min.   : 1.0      Min.   : 2340      Min.   :1780      Min.   : 96.0
## 1st Qu.: 95.0      1st Qu.: 7320      1st Qu.:3597      1st Qu.: 470.0
## Median : 353.0      Median : 9990      Median :4200      Median : 500.0
## Mean   : 855.3      Mean   :10441      Mean   :4358      Mean   : 549.4
## 3rd Qu.: 967.0      3rd Qu.:12925      3rd Qu.:5050      3rd Qu.: 600.0
## Max.   :21836.0     Max.   :21700      Max.   :8124      Max.   :2340.0
##  Personal      PhD      Terminal      S.F.Ratio
##  Min.   : 250      Min.   : 8.00      Min.   : 24.0      Min.   : 2.50
## 1st Qu.: 850      1st Qu.: 62.00      1st Qu.: 71.0      1st Qu.:11.50
## Median :1200      Median : 75.00      Median : 82.0      Median :13.60
## Mean   :1341      Mean   : 72.66      Mean   : 79.7      Mean   :14.09
## 3rd Qu.:1700      3rd Qu.: 85.00      3rd Qu.: 92.0      3rd Qu.:16.50
## Max.   :6800      Max.   :103.00      Max.   :100.0      Max.   :39.80
##  perc.alumni      Expend      Grad.Rate
##  Min.   : 0.00      Min.   : 3186      Min.   : 10.00
## 1st Qu.:13.00      1st Qu.: 6751      1st Qu.: 53.00
## Median :21.00      Median : 8377      Median : 65.00
## Mean   :22.74      Mean   : 9660      Mean   : 65.46
## 3rd Qu.:31.00      3rd Qu.:10830      3rd Qu.: 78.00
## Max.   :64.00      Max.   :56233      Max.   :118.00
```

The variables differ substantially in scale. For example, enrollment and application counts vary widely across schools, as do tuition and instructional expenditures. Graduation rates and percentages such as `Top10perc`, `PhD`, and `perc.alumni` are naturally bounded percentage variables.

3.4 (c-ii) Scatterplot matrix of the first ten variables

```
# check first 10 columns
str(college[, 1:10])
```

```
## 'data.frame': 777 obs. of 10 variables:
## $ : chr "Abilene Christian University" "Adelphi University" "Adrian College" "Agnes Sco
## $ Private : chr "Yes" "Yes" "Yes" "Yes" ...
## $ Apps : int 1660 2186 1428 417 193 587 353 1899 1038 582 ...
## $ Accept : int 1232 1924 1097 349 146 479 340 1720 839 498 ...
## $ Enroll : int 721 512 336 137 55 158 103 489 227 172 ...
## $ Top10perc : int 23 16 22 60 16 38 17 37 30 21 ...
## $ Top25perc : int 52 29 50 89 44 62 45 68 63 44 ...
## $ F.Undergrad: int 2885 2683 1036 510 249 678 416 1594 973 799 ...
## $ P.Undergrad: int 537 1227 99 63 869 41 230 32 306 78 ...
## $ Outstate : int 7440 12280 11250 12960 7560 13500 13290 13868 15595 10468 ...

rownames(college) <- college[, 1]
college <- college[, -1]

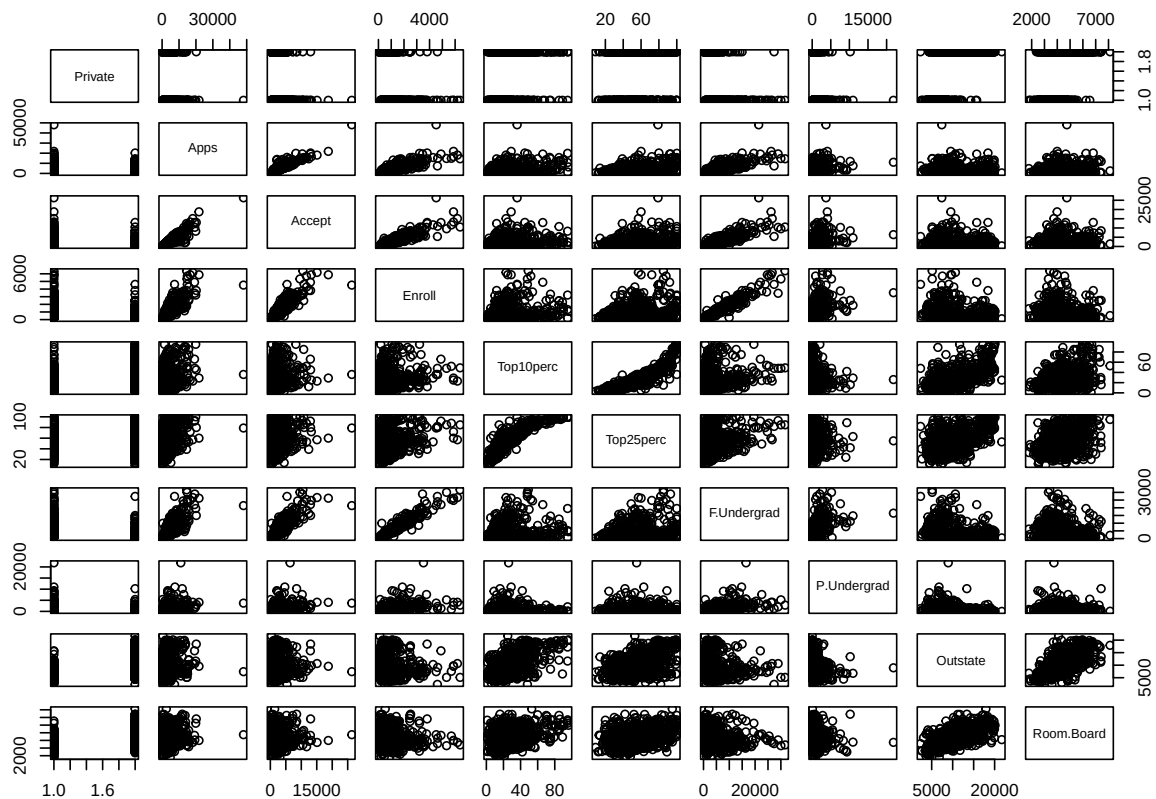
summary(college)
```

```
##      Private      Apps      Accept      Enroll
## Length:777      Min.    :   81      Min.    :   72      Min.    :   35
## Class :character 1st Qu.:  776      1st Qu.:  604      1st Qu.:  242
## Mode  :character Median : 1558      Median : 1110      Median :  434
##                      Mean  : 3002      Mean   : 2019      Mean   :  780
##                      3rd Qu.: 3624      3rd Qu.: 2424      3rd Qu.:  902
##                      Max.   :48094      Max.   :26330      Max.   :6392
##      Top10perc      Top25perc      F.Undergrad      P.Undergrad
## Min.    : 1.00      Min.    :  9.0      Min.    :  139      Min.    :   1.0
## 1st Qu.:15.00      1st Qu.: 41.0      1st Qu.:  992      1st Qu.:  95.0
## Median :23.00      Median : 54.0      Median : 1707      Median :  353.0
## Mean   :27.56      Mean   : 55.8      Mean   : 3700      Mean   :  855.3
## 3rd Qu.:35.00      3rd Qu.: 69.0      3rd Qu.: 4005      3rd Qu.:  967.0
## Max.   :96.00      Max.   :100.0      Max.   :31643      Max.   :21836.0
##      Outstate      Room.Board      Books      Personal
## Min.    : 2340      Min.    :1780      Min.    :  96.0      Min.    :  250
## 1st Qu.: 7320      1st Qu.:3597      1st Qu.: 470.0      1st Qu.:  850
## Median : 9990      Median :4200      Median : 500.0      Median :1200
## Mean   :10441      Mean   :4358      Mean   : 549.4      Mean   :1341
## 3rd Qu.:12925      3rd Qu.:5050      3rd Qu.: 600.0      3rd Qu.:1700
## Max.   :21700      Max.   :8124      Max.   :2340.0      Max.   :6800
##      PhD      Terminal      S.F.Ratio      perc.alumni
## Min.    :  8.00      Min.    : 24.0      Min.    :  2.50      Min.    :  0.00
## 1st Qu.: 62.00      1st Qu.: 71.0      1st Qu.:11.50      1st Qu.:13.00
## Median : 75.00      Median : 82.0      Median :13.60      Median :21.00
## Mean   : 72.66      Mean   : 79.7      Mean   :14.09      Mean   :22.74
## 3rd Qu.: 85.00      3rd Qu.: 92.0      3rd Qu.:16.50      3rd Qu.:31.00
## Max.   :103.00      Max.   :100.0      Max.   :39.80      Max.   :64.00
##      Expend      Grad.Rate
## Min.    : 3186      Min.    : 10.00
## 1st Qu.: 6751      1st Qu.: 53.00
## Median : 8377      Median : 65.00
## Mean   : 9660      Mean   : 65.46
## 3rd Qu.:10830      3rd Qu.: 78.00
## Max.   :56233      Max.   :118.00
```

```
# convert Private to factor before processing
college$Private <- factor(college$Private)

pairs(
  college[, 1:10],
  main = "Scatterplot Matrix: First 10 College Variables"
)
```

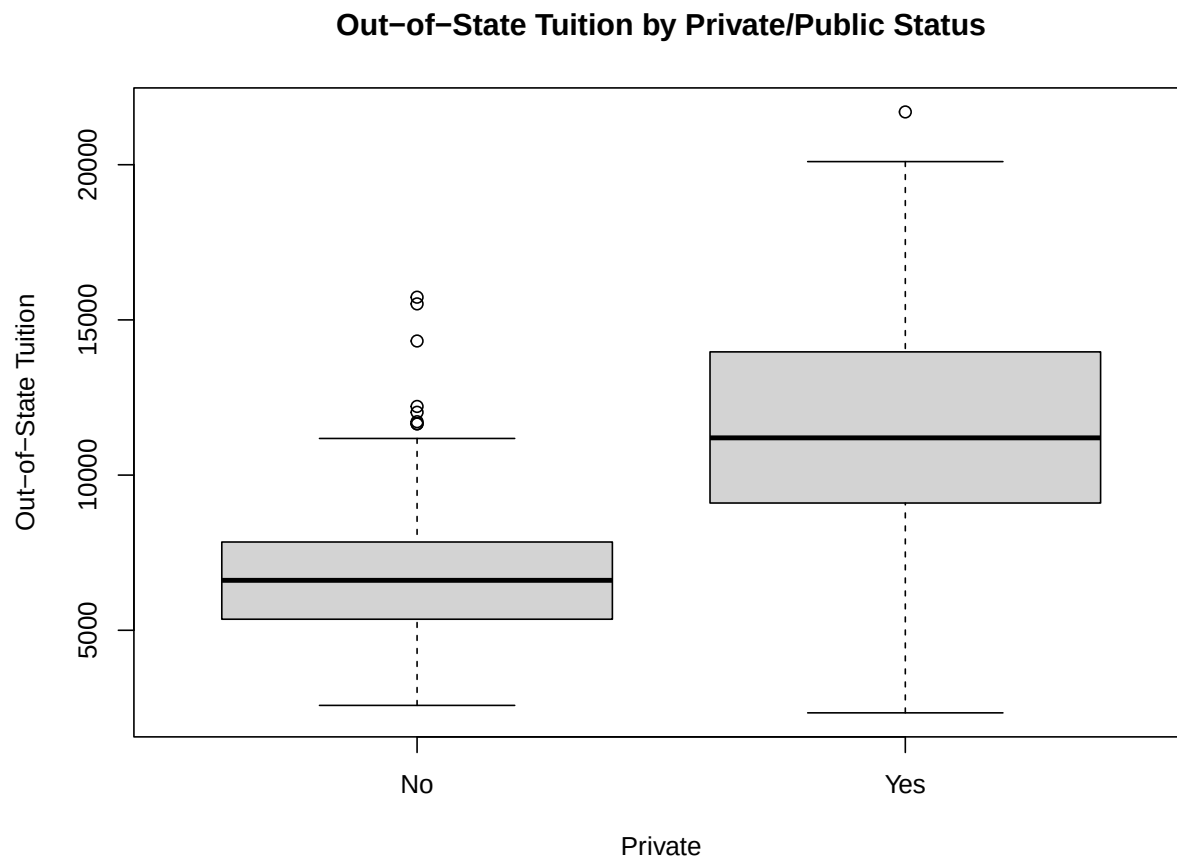
Scatterplot Matrix: First 10 College Variables



Several size-related variables are strongly associated. In particular, colleges receiving more applications also tend to accept and enroll more students. Full-time undergraduate enrollment is also associated with these measures of school size.

3.5 (c-iii) Out-of-state tuition versus private/public status

```
plot(
  college$Private,
  college$Outstate,
  xlab = "Private",
  ylab = "Out-of-State Tuition",
  main = "Out-of-State Tuition by Private/Public Status"
)
```



Private colleges generally have substantially higher out-of-state tuition than public colleges.

3.6 (c-iv) Create an Elite variable

A school is classified as elite when more than 50% of its students come from the top 10% of their high-school class.

```
Elite <- rep("No", nrow(college))
Elite[college$Top10perc > 50] <- "Yes"
Elite <- as.factor(Elite)

college <- data.frame(college, Elite)

summary(college$Elite)
```

```
## No Yes
## 699 78
```

The number of elite colleges is:

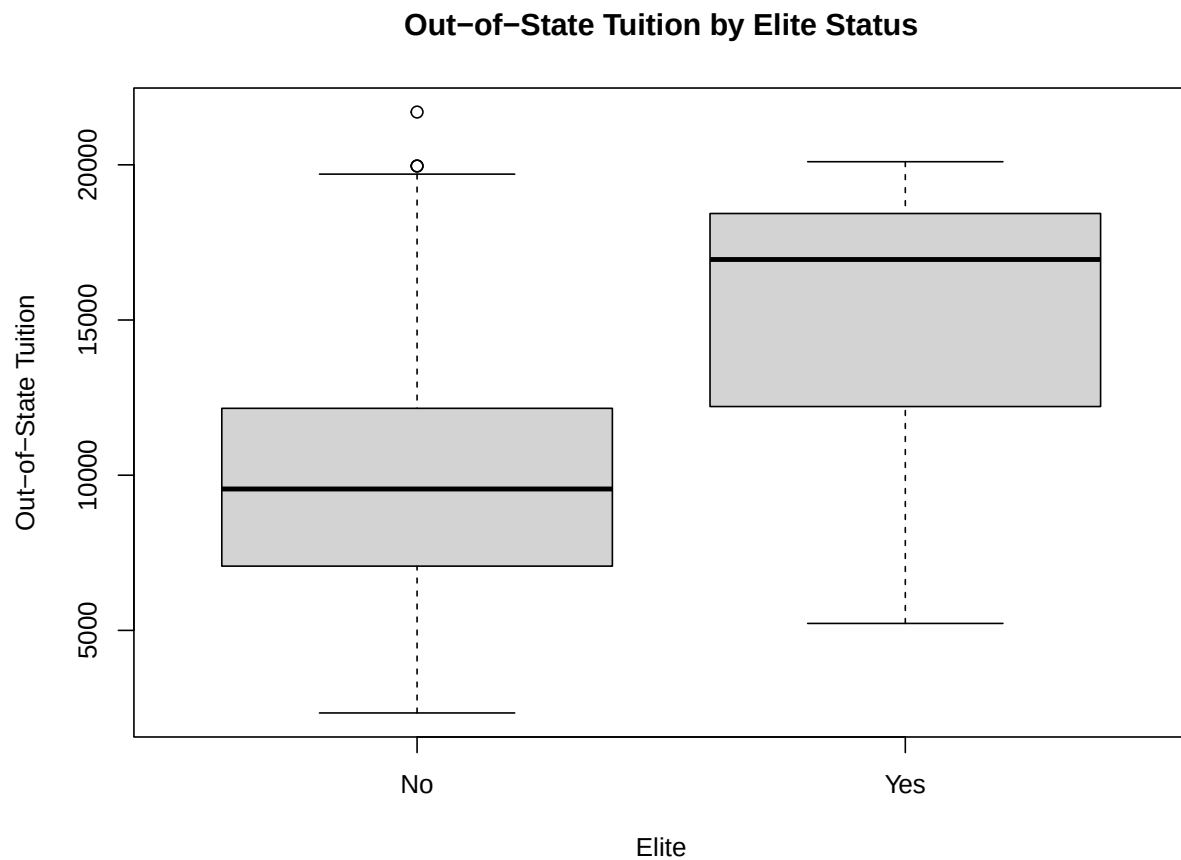
```
sum(college$Elite == "Yes")
```

```
## [1] 78
```

For the standard ISLR College data, this gives **78 elite colleges**.

Now compare tuition between elite and non-elite schools.

```
plot(
  college$Elite,
  college$Outstate,
  xlab = "Elite",
  ylab = "Out-of-State Tuition",
  main = "Out-of-State Tuition by Elite Status"
)
```



Elite colleges tend to have higher out-of-state tuition than non-elite colleges.

3.7 (c-v) Histograms

```
par(mfrow = c(2, 2))

hist(
  college$Apps,
  breaks = 20,
  main = "Applications",
  xlab = "Apps"
)

hist(
  college$Outstate,
```



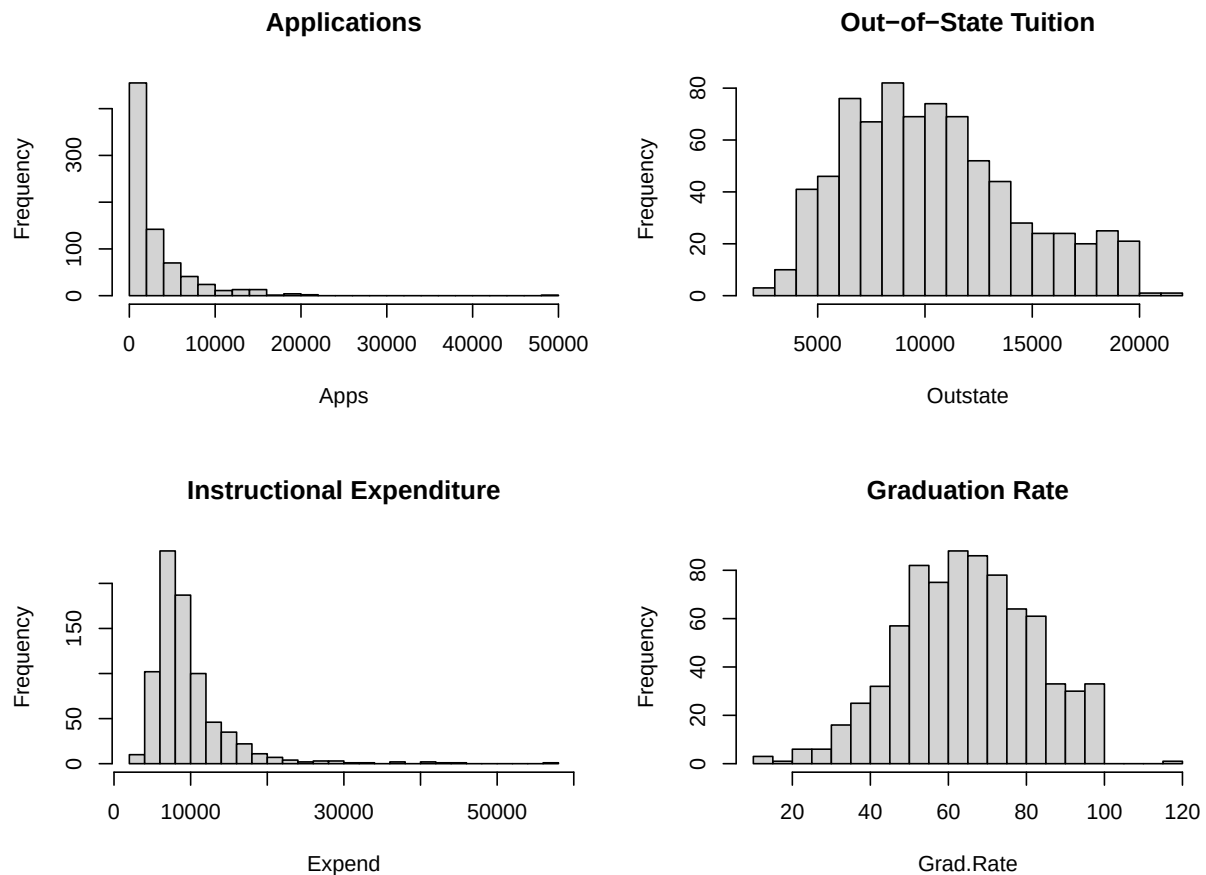
```

breaks = 20,
main = "Out-of-State Tuition",
xlab = "Outstate"
)

hist(
  college$Expend,
  breaks = 20,
  main = "Instructional Expenditure",
  xlab = "Expend"
)

hist(
  college$Grad.Rate,
  breaks = 20,
  main = "Graduation Rate",
  xlab = "Grad.Rate"
)

```



```
par(mfrow = c(1, 1))
```

Applications and instructional expenditure show noticeable right-skewness, reflecting a relatively small number of very large or high-spending institutions. Graduation rates are more concentrated toward the upper part of their possible range.

3.8 (c-vi) Additional exploration

```
# Correlations among selected quantitative variables
```

```
selected <- college[, c(  
  "Apps", "Accept", "Enroll", "Top10perc",  
  "Outstate", "Room.Board", "PhD",  
  "Expend", "Grad.Rate"  
)]
```

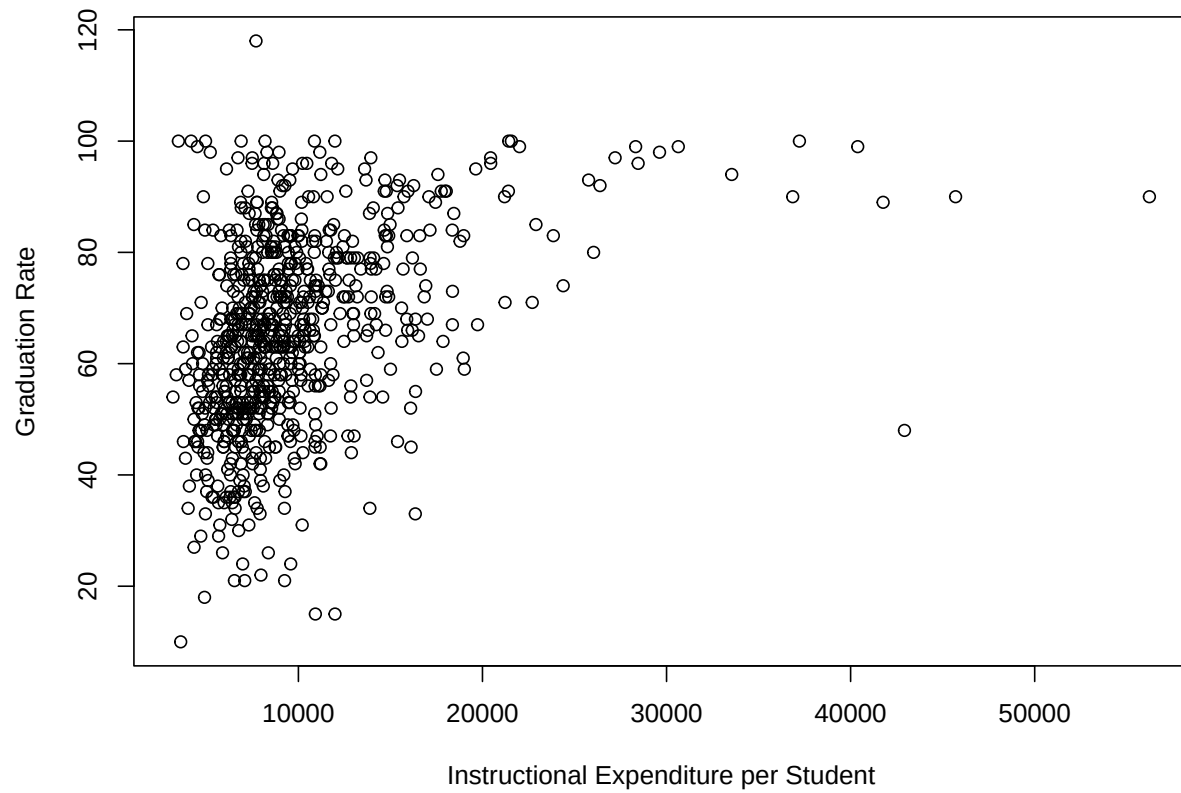
```
round(cor(selected), 2)
```

```
##           Apps Accept Enroll Top10perc Outstate Room.Board  PhD Expend  
## Apps      1.00   0.94   0.85     0.34    0.05     0.16 0.39   0.26  
## Accept    0.94   1.00   0.91     0.19   -0.03     0.09 0.36   0.12  
## Enroll    0.85   0.91   1.00     0.18   -0.16    -0.04 0.33   0.06  
## Top10perc 0.34   0.19   0.18     1.00    0.56     0.37 0.53   0.66  
## Outstate  0.05  -0.03  -0.16     0.56    1.00     0.65 0.38   0.67  
## Room.Board 0.16   0.09  -0.04     0.37    0.65     1.00 0.33   0.50  
## PhD       0.39   0.36   0.33     0.53    0.38     0.33 1.00   0.43  
## Expend    0.26   0.12   0.06     0.66    0.67     0.50 0.43   1.00  
## Grad.Rate 0.15   0.07  -0.02     0.49    0.57     0.42 0.31   0.39  
##           Grad.Rate  
## Apps      0.15  
## Accept    0.07  
## Enroll    -0.02  
## Top10perc 0.49  
## Outstate  0.57  
## Room.Board 0.42  
## PhD       0.31  
## Expend    0.39  
## Grad.Rate 1.00
```

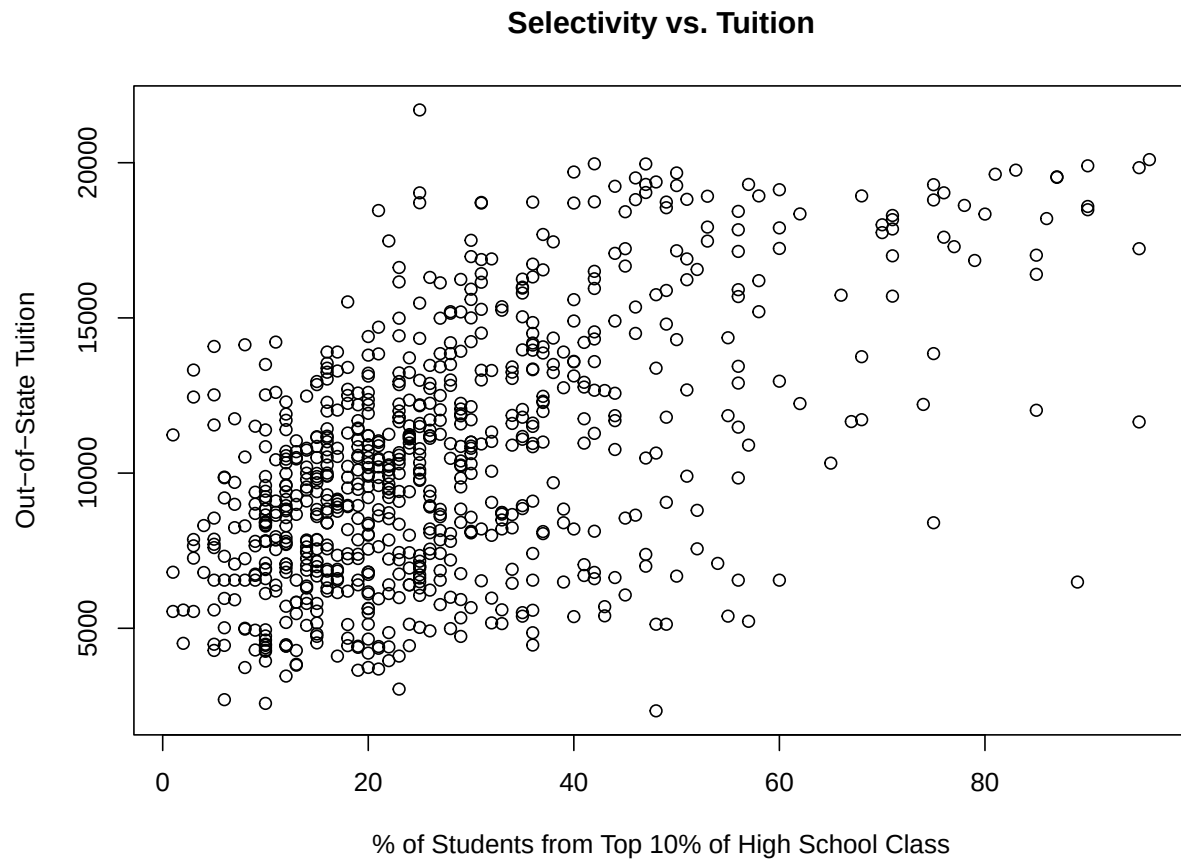
```
# Graduation rate versus instructional expenditure
```

```
plot(  
  college$Expend,  
  college$Grad.Rate,  
  xlab = "Instructional Expenditure per Student",  
  ylab = "Graduation Rate",  
  main = "Graduation Rate vs. Expenditure"  
)
```

Graduation Rate vs. Expenditure



```
# Top 10% students versus tuition
plot(
  college$Top10perc,
  college$Outstate,
  xlab = "% of Students from Top 10% of High School Class",
  ylab = "Out-of-State Tuition",
  main = "Selectivity vs. Tuition"
)
```



Overall, the data suggest strong relationships among measures of institution size. More selective colleges also tend to charge higher tuition, and private or elite colleges generally have higher out-of-state tuition.

4 Exercise 10 - Boston Housing Data

4.1 Setup

```
if (!requireNamespace("ISLR2", quietly = TRUE)) {
  install.packages("ISLR2")
}

library(ISLR2)
data("Boston")
```

4.2 (a) Dimensions and meaning of rows and columns

```
?Boston

dim(Boston)
```

```
## [1] 506 13
```

```
names(Boston)
```

```
## [1] "crim"    "zn"      "indus"   "chas"    "nox"     "rm"      "age"
## [8] "dis"     "rad"     "tax"     "ptratio" "lstat"   "medv"
```

```
str(Boston)
```

```
## 'data.frame':    506 obs. of  13 variables:
## $ crim   : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
## $ zn     : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
## $ indus  : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 ...
## $ chas   : int   0 0 0 0 0 0 0 0 0 ...
## $ nox    : num  0.538 0.469 0.469 0.458 0.458 0.524 0.524 0.524 ...
## $ rm     : num  6.58 6.42 7.18 7 7.15 ...
## $ age    : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
## $ dis    : num  4.09 4.97 4.97 6.06 6.06 ...
## $ rad    : int   1 2 2 3 3 3 5 5 5 ...
## $ tax    : num  296 242 242 222 222 311 311 311 311 ...
## $ ptratio: num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 ...
## $ lstat  : num  4.98 9.14 4.03 2.94 5.33 ...
## $ medv   : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

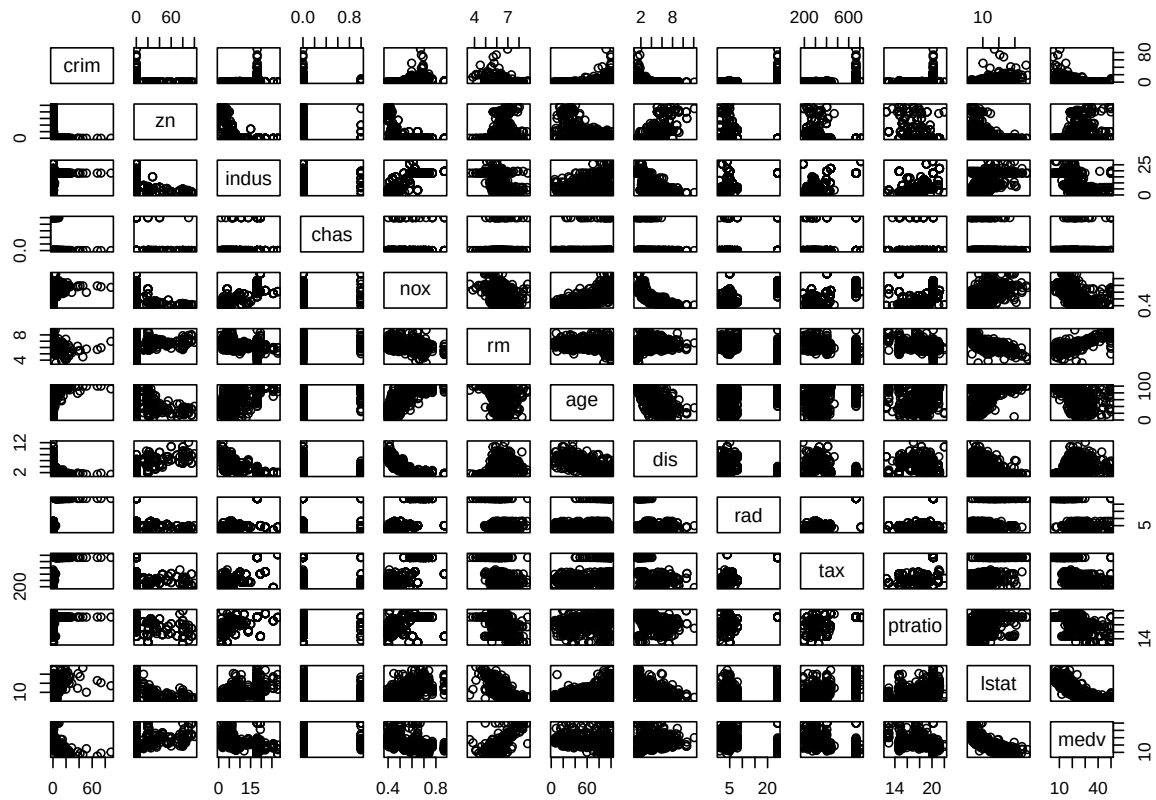
The Boston data contain **506 rows and 13 columns**.

Each row represents a Boston-area census tract. The columns contain tract-level predictors such as crime rate, zoning, industrial land use, air pollution, average number of rooms, property tax rate, pupil-teacher ratio, and related variables. The `medv` variable is the median value of owner-occupied homes.

4.3 (b) Pairwise scatterplots

```
pairs(
  Boston,
  main = "Boston Housing Data: Pairwise Scatterplots"
)
```

Boston Housing Data: Pairwise Scatterplots



Some clear relationships are visible. For example, `medv` tends to increase with `rm` and decrease with `lstat`. Crime tends to be higher in areas with higher `rad` and `tax`, and lower in areas with larger `dis`.

4.4 (c) Predictors associated with per-capita crime rate

```
crime_cor <- sort(
  cor(Boston)[, "crim"],
  decreasing = TRUE
)

crime_cor

##      crim      rad      tax      lstat      nox      indus
## 1.00000000 0.62550515 0.58276431 0.45562148 0.42097171 0.40658341
##      age      ptratio      chas      zn      rm      dis
## 0.35273425 0.28994558 -0.05589158 -0.20046922 -0.21924670 -0.37967009
##      medv
## -0.38830461

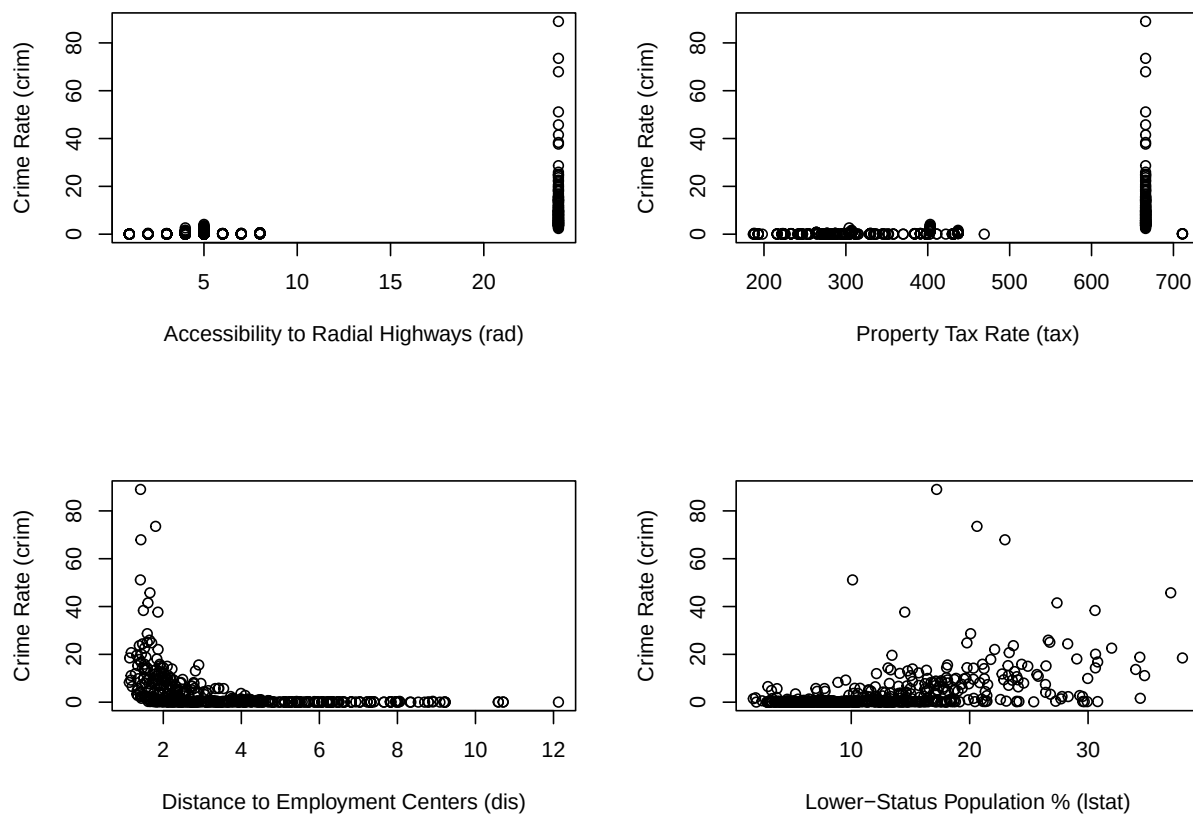
par(mfrow = c(2, 2))

plot(Boston$rad, Boston$crim,
     xlab = "Accessibility to Radial Highways (rad)",
     ylab = "Crime Rate (crim)")
```

```
plot(Boston$tax, Boston$crim,
     xlab = "Property Tax Rate (tax)",
     ylab = "Crime Rate (crim)")

plot(Boston$dis, Boston$crim,
     xlab = "Distance to Employment Centers (dis)",
     ylab = "Crime Rate (crim)")

plot(Boston$lstat, Boston$crim,
     xlab = "Lower-Status Population % (lstat)",
     ylab = "Crime Rate (crim)")
```



```
par(mfrow = c(1, 1))
```

Crime rate is positively associated with variables such as `rad`, `tax`, `indus`, `nox`, and `lstat`, and negatively associated with variables such as `dis`, `rm`, and `medv`. These are associations only and should not be interpreted as causal effects.

4.5 (d) High crime rates, tax rates, and pupil-teacher ratios

```
range(Boston$crim)
```

```
## [1] 0.00632 88.97620
```

```
range(Boston$tax)
```

```
## [1] 187 711
```

```
range(Boston$ptratio)
```

```
## [1] 12.6 22.0
```

```
summary(Boston[, c("crim", "tax", "ptratio")])
```

```
##          crim          tax          ptratio
##  Min.   : 0.00632   Min.   :187.0   Min.    :12.60
## 1st Qu.: 0.08205   1st Qu.:279.0   1st Qu.:17.40
##  Median : 0.25651   Median :330.0   Median :19.05
##  Mean   : 3.61352   Mean    :408.2   Mean    :18.46
## 3rd Qu.: 3.67708   3rd Qu.:666.0   3rd Qu.:20.20
##  Max.   :88.97620   Max.    :711.0   Max.    :22.00
```

There are census tracts with unusually high crime rates relative to the majority of observations. Tax rates and pupil-teacher ratios also show substantial variation, although the extreme skew is especially noticeable for crime.

We can inspect the largest values directly:

```
head(Boston[order(Boston$crim, decreasing = TRUE), ], 10)
```

```
##          crim zn indus chas   nox    rm   age    dis rad tax ptratio lstat medv
## 381 88.9762  0  18.1    0 0.671 6.968  91.9 1.4165  24 666   20.2 17.21 10.4
## 419 73.5341  0  18.1    0 0.679 5.957 100.0 1.8026  24 666   20.2 20.62  8.8
## 406 67.9208  0  18.1    0 0.693 5.683 100.0 1.4254  24 666   20.2 22.98  5.0
## 411 51.1358  0  18.1    0 0.597 5.757 100.0 1.4130  24 666   20.2 10.11 15.0
## 415 45.7461  0  18.1    0 0.693 4.519 100.0 1.6582  24 666   20.2 36.98  7.0
## 405 41.5292  0  18.1    0 0.693 5.531  85.4 1.6074  24 666   20.2 27.38  8.5
## 399 38.3518  0  18.1    0 0.693 5.453 100.0 1.4896  24 666   20.2 30.59  5.0
## 428 37.6619  0  18.1    0 0.679 6.202  78.7 1.8629  24 666   20.2 14.52 10.9
## 414 28.6558  0  18.1    0 0.597 5.155 100.0 1.5894  24 666   20.2 20.08 16.3
## 418 25.9406  0  18.1    0 0.679 5.304  89.1 1.6475  24 666   20.2 26.64 10.4
```

```
head(Boston[order(Boston$tax, decreasing = TRUE), ], 10)
```

```
##          crim zn indus chas   nox    rm   age    dis rad tax ptratio lstat medv
## 489 0.15086  0 27.74    0 0.609 5.454  92.7 1.8209   4 711   20.1 18.06 15.2
## 490 0.18337  0 27.74    0 0.609 5.414  98.3 1.7554   4 711   20.1 23.97  7.0
## 491 0.20746  0 27.74    0 0.609 5.093  98.0 1.8226   4 711   20.1 29.68  8.1
## 492 0.10574  0 27.74    0 0.609 5.983  98.8 1.8681   4 711   20.1 18.07 13.6
## 493 0.11132  0 27.74    0 0.609 5.983  83.5 2.1099   4 711   20.1 13.35 20.1
## 357 8.98296  0 18.10    1 0.770 6.212  97.4 2.1222  24 666   20.2 17.60 17.8
## 358 3.84970  0 18.10    1 0.770 6.395  91.0 2.5052  24 666   20.2 13.27 21.7
## 359 5.20177  0 18.10    1 0.770 6.127  83.4 2.7227  24 666   20.2 11.48 22.7
## 360 4.26131  0 18.10    0 0.770 6.112  81.3 2.5091  24 666   20.2 12.67 22.6
## 361 4.54192  0 18.10    0 0.770 6.398  88.0 2.5182  24 666   20.2  7.79 25.0
```

```
head(Boston[order(Boston$ptratio, decreasing = TRUE), ], 10)
```

```
##          crim zn indus chas   nox    rm   age    dis rad tax ptratio lstat medv
## 355 0.04301 80  1.91    0 0.413 5.663  21.9 10.5857   4 334   22.0  8.05 18.2
## 356 0.10659 80  1.91    0 0.413 5.936  19.5 10.5857   4 334   22.0  5.57 20.6
## 128 0.25915  0 21.89    0 0.624 5.693  96.0  1.7883   4 437   21.2 17.19 16.2
```



```
## 129 0.32543 0 21.89 0 0.624 6.431 98.8 1.8125 4 437 21.2 15.39 18.0
## 130 0.88125 0 21.89 0 0.624 5.637 94.7 1.9799 4 437 21.2 18.34 14.3
## 131 0.34006 0 21.89 0 0.624 6.458 98.9 2.1185 4 437 21.2 12.60 19.2
## 132 1.19294 0 21.89 0 0.624 6.326 97.7 2.2710 4 437 21.2 12.26 19.6
## 133 0.59005 0 21.89 0 0.624 6.372 97.9 2.3274 4 437 21.2 11.12 23.0
## 134 0.32982 0 21.89 0 0.624 5.822 95.4 2.4699 4 437 21.2 15.03 18.4
## 135 0.97617 0 21.89 0 0.624 5.757 98.4 2.3460 4 437 21.2 17.31 15.6
```

4.6 (e) Census tracts bounding the Charles River

The variable `chas` equals 1 for census tracts that bound the Charles River.

```
sum(Boston$chas == 1)
```

```
## [1] 35
```

There are **35 census tracts** that bound the Charles River.

4.7 (f) Median pupil-teacher ratio

```
median(Boston$ptratio)
```

```
## [1] 19.05
```

The median pupil-teacher ratio is **19.05**.

4.8 (g) Census tract with the lowest median home value

```
min_medv <- min(Boston$medv)
min_medv
```

```
## [1] 5
```

```
lowest_rows <- which(Boston$medv == min_medv)
lowest_rows
```

```
## [1] 399 406
```

```
Boston[lowest_rows, ]
```

```
##      crim zn indus chas  nox   rm age   dis rad tax ptratio lstat medv
## 399 38.3518 0  18.1    0 0.693 5.453 100 1.4896 24 666    20.2 30.59    5
## 406 67.9208 0  18.1    0 0.693 5.683 100 1.4254 24 666    20.2 22.98    5
```

The minimum value of `medv` is **5**. Several tracts attain this minimum. To compare those observations with the overall ranges:

```
overall_ranges <- sapply(Boston, range)
```

```
overall_ranges
```

```
##      crim  zn indus chas  nox   rm age   dis rad tax ptratio lstat
## [1,] 0.00632 0 0.46    0 0.385 3.561 2.9 1.1296 1 187    12.6 1.73
## [2,] 88.97620 100 27.74    1 0.871 8.780 100.0 12.1265 24 711    22.0 37.97
##      medv
## [1,] 5
## [2,] 50
```

```
Boston[lowest_rows, ]
```

```
##          crim zn indus chas   nox   rm age   dis rad tax ptratio lstat medv
## 399 38.3518  0  18.1    0 0.693 5.453 100 1.4896 24 666    20.2 30.59    5
## 406 67.9208  0  18.1    0 0.693 5.683 100 1.4254 24 666    20.2 22.98    5
```

The lowest-value tracts tend to have characteristics associated with disadvantaged housing markets, including relatively high crime or lower-status population percentages and, in some cases, low average room counts compared with the rest of the data. The exact values are displayed above so they can be compared directly with each variable's overall range.

4.9 (h) Census tracts averaging more than seven or eight rooms

```
sum(Boston$rm > 7)
```

```
## [1] 64
```

```
sum(Boston$rm > 8)
```

```
## [1] 13
```

There are **64** census tracts with more than seven rooms per dwelling and **13** with more than eight rooms per dwelling.

Inspect the tracts with more than eight rooms:

```
rooms_gt8 <- Boston[Boston$rm > 8, ]
rooms_gt8
```

```
##          crim zn indus chas   nox   rm age   dis rad tax ptratio lstat medv
## 98  0.12083  0  2.89    0 0.4450 8.069 76.0 3.4952  2 276    18.0  4.21 38.7
## 164 1.51902  0 19.58    1 0.6050 8.375 93.9 2.1620  5 403    14.7  3.32 50.0
## 205 0.02009 95  2.68    0 0.4161 8.034 31.9 5.1180  4 224    14.7  2.88 50.0
## 225 0.31533  0  6.20    0 0.5040 8.266 78.3 2.8944  8 307    17.4  4.14 44.8
## 226 0.52693  0  6.20    0 0.5040 8.725 83.0 2.8944  8 307    17.4  4.63 50.0
## 227 0.38214  0  6.20    0 0.5040 8.040 86.5 3.2157  8 307    17.4  3.13 37.6
## 233 0.57529  0  6.20    0 0.5070 8.337 73.3 3.8384  8 307    17.4  2.47 41.7
## 234 0.33147  0  6.20    0 0.5070 8.247 70.4 3.6519  8 307    17.4  3.95 48.3
## 254 0.36894 22  5.86    0 0.4310 8.259  8.4 8.9067  7 330    19.1  3.54 42.8
## 258 0.61154 20  3.97    0 0.6470 8.704 86.9 1.8010  5 264    13.0  5.12 50.0
## 263 0.52014 20  3.97    0 0.6470 8.398 91.5 2.2885  5 264    13.0  5.91 48.8
## 268 0.57834 20  3.97    0 0.5750 8.297 67.0 2.4216  5 264    13.0  7.44 50.0
## 365 3.47428  0 18.10    1 0.7180 8.780 82.9 1.9047 24 666    20.2  5.29 21.9
```

```
summary(rooms_gt8)
```

```
##          crim          zn          indus          chas
## Min.   :0.02009   Min.   : 0.00   Min.   : 2.680   Min.   :0.0000
## 1st Qu.:0.33147   1st Qu.: 0.00   1st Qu.: 3.970   1st Qu.:0.0000
## Median :0.52014   Median : 0.00   Median : 6.200   Median :0.0000
## Mean   :0.71880   Mean   :13.62   Mean   : 7.078   Mean   :0.1538
## 3rd Qu.:0.57834   3rd Qu.:20.00   3rd Qu.: 6.200   3rd Qu.:0.0000
## Max.   :3.47428   Max.   :95.00   Max.   :19.580   Max.   :1.0000
##          nox          rm          age          dis
## Min.   :0.4161   Min.   :8.034   Min.   : 8.40   Min.   :1.801
## 1st Qu.:0.5040   1st Qu.:8.247   1st Qu.:70.40   1st Qu.:2.288
## Median :0.5070   Median :8.297   Median :78.30   Median :2.894
```

##	Mean	:0.5392	Mean	:8.349	Mean	:71.54	Mean	:3.430		
##	3rd Qu.	:0.6050	3rd Qu.	:8.398	3rd Qu.	:86.50	3rd Qu.	:3.652		
##	Max.	:0.7180	Max.	:8.780	Max.	:93.90	Max.	:8.907		
##	rad		tax		prratio		lstat		medv	
##	Min.	: 2.000	Min.	:224.0	Min.	:13.00	Min.	:2.47	Min.	:21.9
##	1st Qu.	: 5.000	1st Qu.	:264.0	1st Qu.	:14.70	1st Qu.	:3.32	1st Qu.	:41.7
##	Median	: 7.000	Median	:307.0	Median	:17.40	Median	:4.14	Median	:48.3
##	Mean	: 7.462	Mean	:325.1	Mean	:16.36	Mean	:4.31	Mean	:44.2
##	3rd Qu.	: 8.000	3rd Qu.	:307.0	3rd Qu.	:17.40	3rd Qu.	:5.12	3rd Qu.	:50.0
##	Max.	:24.000	Max.	:666.0	Max.	:20.20	Max.	:7.44	Max.	:50.0

These tracts generally have large homes and relatively high median home values. However, they are not identical on the other predictors, so a large number of rooms by itself does not completely describe neighborhood conditions or housing value.